

JACOB KLIMCZAK

jacob.klimczak@mail.utoronto.ca | +1 (647) 913-8065 | <https://linkedin.com/in/jacob-klimczak>

EDUCATION

University Of Toronto, Computer Science Specialist (ASIP Co-op Stream)

Sept 2021 - Present

- 3.78/4.0 CGPA.
- Dean's List scholar in 2022 & 2023, recipient of University of Toronto Scholar scholarship in 2021, recipient of Ruth Joyce Norton scholarship in 2022, recipient of Isabel Bader scholarship in 2023.
- Won **CLUE Symposium Best Paper in Data & Modeling** in 2025.
- Nominated for the **Computing Research Association Outstanding Undergraduate Researcher** award by the Faculty of Computer Science at the University of Toronto in 2025.
- Placed top 3 (1st place selection pending) in the **Runway Ready** challenge with Dexterra in 2025 for developing a **demand forecasting** and **queuing model** for facility management at Pearson International Airport in Toronto. This resulted in a partnership where Dexterra funded the development of a prototype solution based on our initial model.
- Won **U of T IMI Big Data and AI Competition** in 2025 for developing a bank customer foundation model created with a **contrastive learning** approach; and applying it to **anomaly detection** models to find instances of money laundering. Scotiabank was so impressed with the model, they tested it for internal use.

RESEARCH EXPERIENCE

Fire Lab, Operations Researcher

Sep 2025 - Present

- Studying optimization of air tanker sharing under Prof. Dave Martell.
- Building multi-agent simulation models of air tankers to evaluate and optimize performance of scheduling, dispatch, and sharing policies.
- Architecting and creating **Python** simulation bed to support event-based simulations.

DGP Lab, Machine Learning Researcher

May 2025 - Present

- Studying AI for environmental modeling and visualization under Prof. Ron Baecker in the DGP Lab.
- Building on top of **Microsoft's Perceiver Encoder** and **SWIN Transformer-based Aurora foundation model** to extend its capabilities to cover Canadian fire forecasting.
- Fine-tuning the Aurora model for enhanced fire forecasting capabilities and building inference pipelines to quickly estimate fire risk in any part of the Boreal Forest.
- **Conferences**
 - Gismondi, G., Klimczak, J., Seok, J. (2025). Visualization of Extreme Climate Events. *2025 GIS in Education and Research Conference*.

Freight Data Warehouse, Data Specialist

Jan 2023 - Present

- Providing data support and studying transportation economics and modeling under Prof. Matthew Roorda.
- Researching the use of **genetic algorithms** to solve place-and-route problems in urban planning.
- Acting as technical lead for one of Canada's largest academic transportation data repositories, working with partners such as Transport Canada, the Ministry of Transportation of Ontario, and the American Transportation Research Institute to ensure their data was secure.
- Supporting 20+ cross-disciplinary research projects, partnered with 5 universities, through anonymizing and distributing sensitive data from Postgres, MongoDB, MySQL and Microsoft SQL Server databases.
- Processing **180+ TB** with **Python** and **C# ETL pipelines** for Transport CAN to model emissions on 401.
- Contributed to *Dashboard Visualization of GHGs and Air Contaminants*, an report for Transport Canada on methods for visualizing traffic-related emissions with dashboard tools.
- Collaborating with leading GIS companies to develop visualization prototypes for urban developments.

- **Publications**
 - Wagner, S., Diamond, S., Kossowsky, D., Wang, C., Klimczak, J., and Roorda, M. (2026). A Tool for Visualizing the Physical Elements of Communities. *HCI International*.
 - Rivera-Gonzalez, C., Feng, Z., Klimczak, J., Bayanouni, H., Carr, K., & Roorda, M. (2025). A Freight Data Repository as a Foundational Pillar for Urban Freight Research. *Research in Transportation Business & Management*, 61, 101417. <https://doi.org/10.1016/j.rtbm.2025.101417>
 - Rivera-Gonzalez, C., Klimczak, J., & Roorda, M. (2024). Dashboard Visualization of GHGs and Air Contaminants. *Government Report for Transport Canada*.
- **Under Review**
 - Wagner, S., Klimczak, J., Kossowsky, D., Diamond, S., Roorda, M. (2025). Stated Preference Surveys for Generative Design of Communities. *ISCTSC Procedia*.
 - Wagner, S., Santos, G.M., Klimczak, J., Kossowsky, D., Diamond, S., Roorda, M. (2025). A Generative Design Tool for Designing the Physical Elements of Communities. *Journal of Urban Management*.
- **Conferences**
 - Klimczak, J., Rivera-Gonzalez, C., Roorda, M. (2025). Quantifying the Effects of Induced Demand by the Toronto Blue Jays. *4th City Logistics for the Urban Economy Symposium*.
 - Klimczak, J., Rivera-Gonzalez, C., Roorda, M. (2024). A Smart Freight Dashboard for Airborne Pollutants: Visualizing Emissions in Southern Ontario. *3rd City Logistics for the Urban Economy Symposium*.
 - Klimczak, J., Rivera-Gonzalez, C., Roorda, M. (2024). Gridlock in the Dugout: Induced Demand and the Toronto Blue Jays. *59th Canadian Transportation Research Forum*.
 - Wagner, S., Klimczak, J., Santos, G.M., Kossowsky, D., Diamond, S., Roorda, M. (2024). A Generative Design Tool for Designing Complete Communities. *8th International Conference on Urban E-Planning*.

James Lab, Fire Researcher

Nov 2022 - Present

- Studying wildfire behavior and computational modeling under Prof. Patrick James.
- Researching the use of **signal processing techniques** (wavelets) for analysis of 3D fuel structure to detect insects, characterize fire risk, and summarize TLS scans.
- Building resource-efficient, **cellular-automata**, fire models to allow for quick stand-level fire simulation.
- Creating open-source, parallelized voxelizations tools in **Go**.
- Examining the Canadian Forest Service's TLS guidelines to reduce field costs by eliminating redundant scans from their recommended configuration.
- **In Preparation**
 - Klimczak, J., James, P.M.A., (2026). Using Wavelets and LIDAR to Characterize 3D Forest Fuel Structure. *Planned for Ecological Informatics*
- **Conferences**
 - Dewez, N.P., James, P.M.A., Klimczak, J. (2024). Effects of Spruce Budworm Defoliation Legacies on Ignitions in the Eastern Boreal Forest. *2024 Wildland Fire Canada Conference*.

WORK EXPERIENCE

Intel/Altera, Software Engineer Intern

May 2024 - Sep 2025

- Contributed features to a C++ and Qt-based power modeling application for **FPGAs**.
- Designed and implemented index data structures to ensure data about any portion of a 10GB+ chip design was instantly accessible through visual and command-line tools.
- Tracked and solved **memory corruption bugs** introduced through external dependencies.
- Designed and implemented workarounds to limitations of large open-source libraries (Qt).
- Stood up new test suites to confirm power modeling results matched ground truth with new models.
- **Modeled power consumption** for Agilix family FPGAs.

Temerity Analytics Inc., Full-Stack Developer*May 2023 - Feb 2024*

- Optimized and created components for **Angular** frontend and **Django** backend.
- Led expansion of backend to support new mobile application for tracking vehicles.
- Architected and built data solutions in **PostgreSQL** to compute vehicle statistics in real time.

Rotogran International Inc., Full-Stack Developer*Nov 2019 - Jan 2023*

- Contributed to **C#** & **Python** backends. Wrote proprietary communication protocol on top of TCP stack.
- Developed multiple **C#** and **Java** SDKs for API consumers to interact with proprietary and REST APIs.
- Worked on web platforms in **React**, and desktop and mobile applications with **C#** and **Java/Android** SDKs.
- Architected **MSSQL** database, and wrote **C#** pipelines to digitize existing paper records.
- Containerized existing backends with **Docker** for easier CI/CD.